Existence of a Point for an Integral Identity

Topic.

Definite integrals, continuity, Rolle's theorem.

Statement.

Let f be continuous on $[a,b]$, where $a>0$, and suppose that

$$\int_a^b f(x)\,dx = 0.$$

Prove that there exists a point

$$c \in (a,b)$$

such that

$$\int_a^c f(x)\,dx = cf(c).$$

Solution.

Let

$$F(x) = \int_a^x f(t)\,dt.$$

Then F is differentiable on (a,b) and

$$F'(x) = f(x).$$

Since $a>0$, consider the function

$$\varphi(x) = \frac{F(x)}{x}, \qquad x \in [a,b].$$

The function φ is continuous on $[a,b]$ and differentiable on (a,b).

We have

$$\varphi(a) = \frac{F(a)}{a} = \frac{1}{a}\int_a^a f(x)\,dx = 0.$$

Also,

$$\varphi(b) = \frac{F(b)}{b} = \frac{1}{b}\int_a^b f(x)\,dx = 0.$$

Therefore,

$$\varphi(a) = \varphi(b).$$

By Rolle's theorem, there exists a point

$$c \in (a, b)$$

such that

$$\varphi'(c) = 0.$$

Now compute the derivative:

$$\varphi'(x) = \left(\frac{F(x)}{x}\right)' = \frac{xF'(x) - F(x)}{x^2}.$$

Therefore,

$$\varphi'(c) = 0$$

implies

$$\frac{cF'(c) - F(c)}{c^2} = 0.$$

Hence,

$$cF'(c) = F(c).$$

Substitute

$$F'(c) = f(c)$$

and

$$F(c) = \int_a^c f(x)\,dx.$$

Then we get

$$cf(c) = \int_a^c f(x)\,dx.$$

Therefore,

$$\int_a^c f(x)\,dx = cf(c).$$